6 b output.sci (C:\Users\LENOVO\Documents\6 b output.sci) - SciNotes

*6 b output.sci

```
// Test Case: Generation of Periodic and Aperiodic Signals
// Parameters for the periodic signal (Sinusoidal)
f_periodic = 1; // Frequency in Hz
A_periodic = 1; // Amplitude
t_periodic = 0:0.01:2; // Time vector for 0 to 2 seconds
// Generate the periodic signal
x_periodic = A_periodic * sin(2 * %pi * f_periodic * t_periodic);
// Plot the periodic signal
clf; // Clear the current figure
subplot(2,1,1); // Subplot 1 for the periodic signal
plot(t_periodic, x_periodic);
xlabel('Time (s)');
ylabel('Amplitude');
title('Periodic Signal: Sinusoidal Wave (1 Hz)');
// Parameters for the aperiodic signal (Exponentially Decaying)
a_aperiodic = 0.5; // Base of the exponential decay
A_aperiodic = 2; // Amplitude
t_aperiodic = 0:0.01:2; // Time vector for 0 to 2 seconds
// Generate the aperiodic signal
x_aperiodic = A_aperiodic * exp(-a_aperiodic * t_aperiodic);
// Plot the aperiodic signal
subplot(2,1,2); // Subplot 2 for the aperiodic signal
plot(t_aperiodic, x_aperiodic);
xlabel('Time (s)');
ylabel('Amplitude');
title('Aperiodic Signal: Exponentially Decaying Signal');
```

Graphic window number 1

File Tools Edit ?

